

# PROS1-A690-OSB-M

Serie PROS-M · R1234ZE · senza serbatoio / without tank



PROS1-A690-OSB-M

RESA FRIGORIFERA · COOLING CAPACITY

**446** kW

EER · 12/7 °C · AIR 35 °C

**2,98**

RESA NOMINALE · 15 °C · AIR 25 °C

**684** kW

PORTATA ACQUA · WATER FLOW

**76,8** m<sup>3</sup>/h

|                            |                 |                                        |                       |
|----------------------------|-----------------|----------------------------------------|-----------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 7.015 × 2.197 × 2.442 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                     |
| Ventilatori / Fans         | 12              | Taglia / Size                          | 690                   |
| Refrigerante / Refrigerant | R1234ZE · GWP 7 | Campo aria esterna / Ambient air range | -5 / +50 °C           |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C | 50 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 502   | 474   | 446   | 418   | 390   | 361   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 684   | 648   | 612   | 575   | 538   | 500   |

Refrigeratori condensati ad aria con compressori a vite e condensatore a microcanali, selezionabili con acqua in uscita fino a 17 °C: indicati per i processi che lavorano con acqua non troppo fredda.

Air-cooled chillers with screw compressors and micro-channel condenser, selectable with leaving water up to 17 °C: suited to processes that run on water that is not too cold.

# Dati tecnici

## Technical data

PROS1-A690-0SB-M

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 683,98 kW                |
| Resa frigorifera / Cooling capacity | 588.223 kcal/h           |
| Potenza assorbita / Power input     | 128,69 kW                |
| EER / EER                           | 5,32                     |
| Portata acqua / Water flow          | 117,64 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                         |
|-------------------------------------|-------------------------|
| Resa frigorifera / Cooling capacity | 446,28 kW               |
| Potenza assorbita / Power input     | 149,96 kW               |
| EER / EER                           | 2,98                    |
| ESEER / ESEER                       | 3,96                    |
| Portata acqua / Water flow          | 76,76 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                       |
|---------------------------------------------|-----------------------|
| Compressori / Compressors                   | 2 × Vite              |
| Circuiti frigoriferi / Refrigerant circuits | 2                     |
| Ventilatori / Fans                          | 12                    |
| Condensatore / Condenser                    | a microcanali         |
| Evaporatore / Evaporator                    | a fascio tubiero      |
| Refrigerante / Refrigerant                  | R1234ZE · GWP 7 · A2L |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 7.015 × 2.197 × 2.442 mm  |
| Campo aria esterna / Ambient air range                                                    | -5 / +50 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 · 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 437,4 kW     |
| MEG 30 % | × 0,950 | 424,0 kW     |
| MEG 40 % | × 0,920 | 410,6 kW     |
| MPG 20 % | × 0,975 | 435,1 kW     |
| MPG 30 % | × 0,960 | 428,4 kW     |
| MPG 40 % | × 0,950 | 424,0 kW     |

# Prestazioni Performance

PROS1-A690-OSB-M · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 434 | 413 | 391 | 369 | 347 | 324 | 301 | 278 |
| 2             | 453 | 431 | 408 | 385 | 362 | 339 | 315 | 291 |
| 3             | 472 | 449 | 426 | 402 | 378 | 353 | 329 | 304 |
| 4             | 492 | 468 | 444 | 419 | 394 | 369 | 343 | 318 |
| 5             | 512 | 487 | 462 | 437 | 411 | 385 | 358 | 331 |
| 6             | 533 | 508 | 482 | 455 | 429 | 401 | 374 | 346 |
| 7             | 554 | 528 | 502 | 474 | 446 | 418 | 390 | 361 |
| 8             | 577 | 550 | 522 | 494 | 465 | 436 | 407 | 377 |
| 9             | 600 | 571 | 543 | 514 | 484 | 454 | 423 | 392 |
| 10            | 624 | 595 | 565 | 535 | 504 | 473 | 441 | 409 |
| 11            | 648 | 618 | 587 | 556 | 524 | 492 | 459 | 426 |
| 12            | 674 | 642 | 611 | 578 | 545 | 512 | 478 | 444 |
| 13            | 699 | 667 | 634 | 600 | 567 | 532 | 497 | 462 |
| 14            | 726 | 693 | 659 | 624 | 589 | 554 | 517 | 481 |
| 15            | 753 | 719 | 684 | 648 | 612 | 575 | 538 | 500 |
| 16            | 782 | 746 | 710 | 673 | 636 | 598 | 559 | 520 |
| 17            | 811 | 774 | 737 | 698 | 660 | 620 | 581 | 541 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 104 | 113 | 123 | 135 | 147 | 161 | 176 | 193 |
| 2             | 104 | 114 | 124 | 135 | 148 | 162 | 177 | 194 |
| 3             | 105 | 114 | 124 | 136 | 148 | 162 | 178 | 194 |
| 4             | 105 | 114 | 125 | 136 | 149 | 163 | 178 | 195 |
| 5             | 105 | 115 | 125 | 136 | 149 | 163 | 179 | 196 |
| 6             | 106 | 115 | 125 | 137 | 150 | 164 | 179 | 196 |
| 7             | 106 | 115 | 126 | 137 | 150 | 164 | 180 | 197 |
| 8             | 106 | 116 | 126 | 137 | 150 | 165 | 180 | 197 |
| 9             | 107 | 116 | 126 | 138 | 151 | 165 | 181 | 198 |
| 10            | 107 | 116 | 127 | 138 | 151 | 165 | 181 | 198 |
| 11            | 108 | 117 | 127 | 139 | 151 | 166 | 182 | 199 |
| 12            | 108 | 117 | 127 | 139 | 152 | 166 | 182 | 199 |
| 13            | 108 | 118 | 128 | 139 | 152 | 167 | 182 | 200 |
| 14            | 109 | 118 | 128 | 140 | 153 | 167 | 183 | 200 |
| 15            | 109 | 118 | 129 | 140 | 153 | 167 | 183 | 201 |
| 16            | 110 | 119 | 129 | 140 | 153 | 168 | 184 | 201 |
| 17            | 110 | 119 | 129 | 141 | 154 | 168 | 184 | 201 |

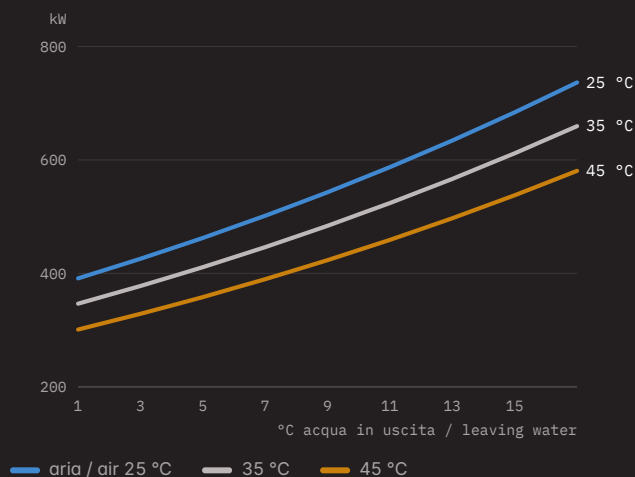
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   | 50   |
|---------------|------|------|------|------|------|------|------|------|
| 1             | 4,18 | 3,65 | 3,17 | 2,74 | 2,35 | 2,01 | 1,71 | 1,44 |
| 2             | 4,35 | 3,79 | 3,30 | 2,85 | 2,45 | 2,09 | 1,78 | 1,50 |
| 3             | 4,51 | 3,94 | 3,43 | 2,96 | 2,55 | 2,18 | 1,85 | 1,56 |
| 4             | 4,69 | 4,09 | 3,56 | 3,08 | 2,65 | 2,27 | 1,93 | 1,63 |
| 5             | 4,86 | 4,25 | 3,70 | 3,20 | 2,76 | 2,36 | 2,00 | 1,69 |
| 6             | 5,05 | 4,41 | 3,85 | 3,33 | 2,87 | 2,45 | 2,09 | 1,76 |
| 7             | 5,23 | 4,58 | 3,99 | 3,46 | 2,98 | 2,55 | 2,17 | 1,83 |
| 8             | 5,42 | 4,75 | 4,14 | 3,59 | 3,09 | 2,65 | 2,26 | 1,91 |
| 9             | 5,62 | 4,92 | 4,30 | 3,73 | 3,21 | 2,75 | 2,34 | 1,98 |
| 10            | 5,82 | 5,11 | 4,46 | 3,87 | 3,34 | 2,86 | 2,44 | 2,06 |
| 11            | 6,02 | 5,29 | 4,62 | 4,01 | 3,46 | 2,97 | 2,53 | 2,14 |
| 12            | 6,24 | 5,48 | 4,79 | 4,16 | 3,59 | 3,08 | 2,63 | 2,23 |
| 13            | 6,45 | 5,67 | 4,96 | 4,31 | 3,72 | 3,20 | 2,73 | 2,31 |
| 14            | 6,67 | 5,87 | 5,14 | 4,47 | 3,86 | 3,32 | 2,83 | 2,40 |
| 15            | 6,89 | 6,07 | 5,32 | 4,63 | 4,00 | 3,44 | 2,94 | 2,49 |
| 16            | 7,12 | 6,27 | 5,50 | 4,79 | 4,15 | 3,56 | 3,05 | 2,59 |
| 17            | 7,34 | 6,48 | 5,69 | 4,96 | 4,29 | 3,69 | 3,16 | 2,68 |

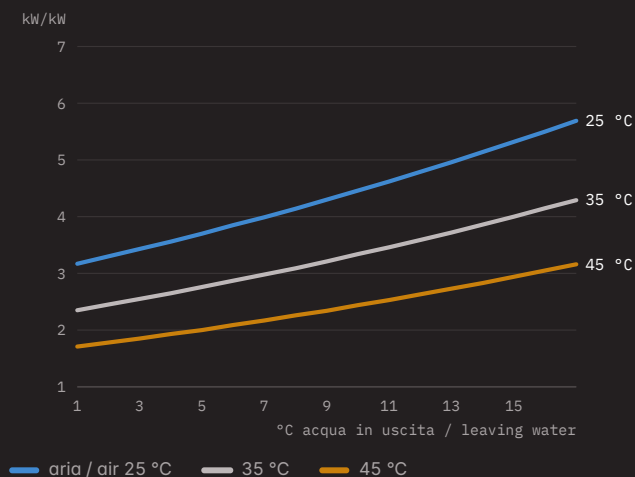
# Curve e dimensioni

## Curves and dimensions

### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 7.015 × 2.197 × 2.442 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

PROS1-A690-OSB-M · 66 optional disponibili per questa configurazione · 66 options available for this configuration

## Pompe / Pumps

|                                                                                                    |
|----------------------------------------------------------------------------------------------------|
| Pompa 92 m³/h – prevalenza 25 m<br>Pump 92 m³/h – head 25 m                                        |
| Pompa 92 m³/h – prevalenza 40 m<br>Pump 92 m³/h – head 40 m                                        |
| Pompa doppia montata 92 m³/h – prevalenza 25 m<br>Twin pump, factory - fitted, 92 m³/h – head 25 m |
| Pompa doppia montata 92 m³/h – prevalenza 40 m<br>Twin pump, factory - fitted, 92 m³/h – head 40 m |

## Serbatoi e vasi di espansione / Tanks and expansion vessels

|                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia montata<br>External water tank 2000 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 1500 l, AISI 304 with factory - fitted single AISI 304 pump |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory - fitted single AISI 304 pump |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory - fitted single AISI 304 pump |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory - fitted single AISI 304 pump |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola montata<br>External water tank 1500 l, AISI 304 with factory - fitted single pump                      |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata<br>External water tank 2000 l, AISI 304 with factory - fitted single pump                      |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory - fitted single pump                      |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory - fitted single pump                      |

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 2000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 2000 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 3000 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 5000 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 1500 l, pressurised, galvanised steel with factory - fitted single pump |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory - fitted single pump |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 5000 l, pressurised, galvanised steel with factory - fitted single pump |
| Vaso di espansione 100 l<br>Expansion vessel 100 l                                                                                                                                           |
| Vaso di espansione 200 l<br>Expansion vessel 200 l                                                                                                                                           |
| Vaso di espansione 300 l<br>Expansion vessel 300 l                                                                                                                                           |
| Vaso di espansione 500 l<br>Expansion vessel 500 l                                                                                                                                           |
| Vaso di espansione 750 l<br>Expansion vessel 750 l                                                                                                                                           |

## Valvole e filtri / Valves and filters

|                                                       |
|-------------------------------------------------------|
| Elettrovalvola 2"<br>Solenoid valve 2"                |
| Elettrovalvola 3"<br>Solenoid valve 3"                |
| Elettrovalvola 4"<br>Solenoid valve 4"                |
| Elettrovalvola 5"<br>Solenoid valve 5"                |
| Elettrovalvola 6"<br>Solenoid valve 6"                |
| Filtro a dischi in acciaio 2"<br>Steel disc filter 2" |

# Optional — 2/2

## Options

PROS1-A690-OSB-M · 66 optional disponibili per questa configurazione · 66 options available for this configuration

### Valvole e filtri / Valves and filters

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y-strainer 2"

Filtro meccanico a Y 3"

Y-strainer 3"

Filtro meccanico a Y 4"

Y-strainer 4"

Filtro meccanico a Y 5"

Y-strainer 5"

Filtro meccanico a Y 6"

Y-strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

### Free cooling / Free cooling

Free cooling 500 kW – 12 ventilatori (rif. 8)

Free cooling 500 kW – 12 fans (ref. 8)

### Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
- Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
- Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
- Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
- Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
- With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
- Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
- Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
- Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

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Continuity by design